

Chapter 6

What-If Analysis

Using Decision-Making Tools

Summary of problem:

- Assume you want to buy your first home.
- After doing some preliminary research on prices, you developed a spreadsheet to help you calculate your monthly mortgage payment, total amount to repay the loan, and the total amount of interest you will pay.
- Your total budget for the home is \$150,000 including taxes, closing costs, and other miscellaneous fees.
- You plan to take \$10,000 out of your savings account for a down payment.
- You are currently investigating loan interest rates at various banks and credit unions.
- You realize that you may need to find a less expensive home or increase your down payment to reach a monthly payment you can afford.
- Although you can change input values to see how different values affect the monthly payment, you want to be able to see the comparisons at the same time. In addition, you want to look at your budget to review the impact of purchasing a new home on your income and expenses.
- To help you make a decision, you will use several what-if analysis tools, each with specific purposes, benefits, and restrictions.

Part (1)

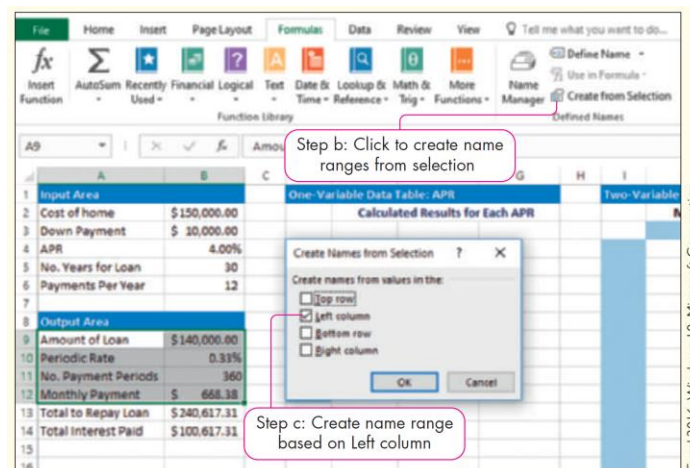
Summary:

You decide to simplify the PMT function you are using to calculate your mortgage by adding range names for the Amount of Loan, Periodic Rate, and No. of Payments instead of the actual cell references. After creating the range names, you will modify the names and create a list of range names.

Step 1: CREATE A RANGE NAME

You want to assign a range name to the Amount of Loan, Periodic Rate, and No. of Payments.

- a. Open e06h1Mortgage and save it as e06h1Mortgage_LastName_FirstName.
- b. Select the range **A9:B12**, click the **Formulas** tab, select **Create from Selection** in the Defined Names group, and then ensure **Left column** is checked. This automatically assigns range names to values based on the row labels in Column A.
- c. Click OK in the Create Names from Selection dialog box.
Range names are created for cells B9:B12 and when any cell in that range is selected, the name will appear in Name Box.
- d. Save the workbook.

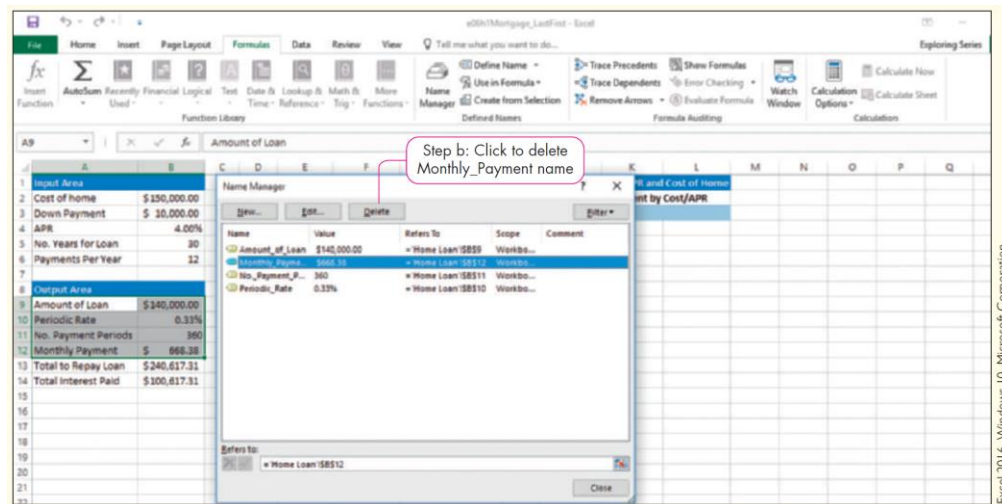


Step 2: EDIT AND DELETE RANGE NAMES

You noticed that an extra row was selected when the name ranges were created. You will use the Name Manager dialog box to view and delete the unneeded name range.

a. Click **Name Manager** in the Defined Names group on the Formulas tab.

b. Select **Monthly_Payment** in the list of named ranges and click **Delete**, read the warning message box, and then click **OK** to confirm the deletion of the Monthly_Payment range name. Click **Close** and Save the workbook.



STEP 3: USE A RANGE NAME IN A FORMULA

You will modify the PMT function by replacing the existing references with the corresponding range names. This will help interpret the PMT function.

- Click **cell B12**, the cell containing the PMT function.
- Click **Insert Function** (between the Name Box and the Formula Bar) to open the Function Arguments dialog box.
- Click in the **Rate box**, click **cell B10**, and then press **Tab**. Click **cell B11**, press **Tab**, and type **-**, and then click **B9**. The new function is **=PMT(Periodic_Rate,No._Payment_Periods,Amount_of_Loan)**.
- Click **OK**. Save the workbook.

The screenshot shows the Excel interface with the **Formulas** tab selected. The **Function Library** group contains the **Insert Function** button. The **Formula Bar** displays the formula **=PMT(Periodic_Rate,No._Payment_Periods,Amount_of_Loan)**. The **Function Arguments** dialog box for the **PMT** function is open, showing the following arguments:

Argument	Value	Result
Rate	Periodic_Rate	= 0.003333333
Nper	No._Payment_Periods	= 360
Pv	-Amount_of_Loan	= -140000
Fv		= number
Type		= number

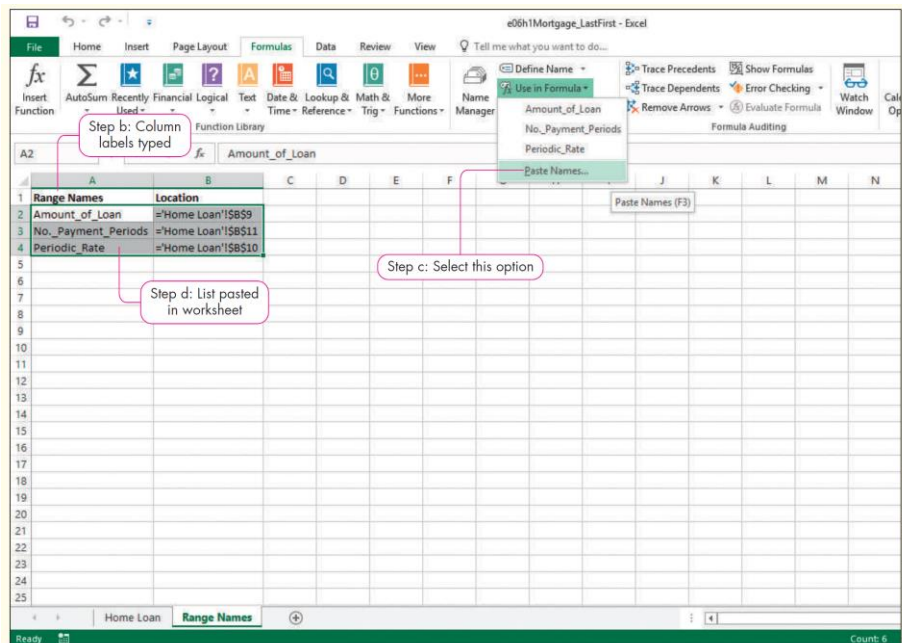
The **Formula result** is **\$668.38**. The **Formula Bar** shows the formula **=PMT(Periodic_Rate,No._Payment_Periods,Amount_of_Loan)**. The **Function Arguments** dialog box also includes a description: "Calculates the payment for a loan based on constant payments and a constant interest rate. Rate is the interest rate per period for the loan. For example, use 6%/4 for quarterly payments at 6% APR."

Annotations in the image indicate the steps: "Step b: Click Insert Function" points to the **Insert Function** button; "Step c: Ranges names in formula in the Formula Bar" points to the formula in the **Formula Bar**; and "Step c: Ranges names entered as arguments in dialog box" points to the arguments in the **Function Arguments** dialog box.

STEP 4: INSERT A LIST OF RANGE NAMES

To ensure continuity moving forward, you want to create a documentation worksheet that lists all of the range names in the workbook.

- a. Click **New sheet** to the right of the Home Loan tab and double-click the default sheet name **Sheet1**. Type **Range Names** and press **Enter**. You inserted and renamed the new worksheet to reflect the data you will add to it.
- b. Type **Range Names** in **cell A1** and type **Location** in **cell B1**. Bold these headings. These column headings will display above the list of range names.
- c. Click **cell A2**, click **Use in Formula** in the Defined Names group on the Formulas tab, and then select **Paste Names**. The Paste Name dialog box opens, displaying all of the range names in the workbook.
- d. Click **Paste List**. Excel pastes an alphabetical list of range names starting in cell A2. The second column displays the locations of the range names.
- e. Increase the widths of columns A and B to fit the data.
- f. Save the workbook.



Part (2)

Summary:

As you consider different options for a home purchase, you want to use data tables to compare how different interest rates and prices will affect your monthly payment. You decide to create both one- and two-variable data tables to analyze the results.

Step 1: SET UP ONE-VARIABLE SUBSTITUTION VALUES

You want to compare monthly mortgage payments, total amounts to repay a loan, and total interest you will pay based on several interest rates—the variable. The interest rates range from 4% to 6% in 0.25% increments. Your first step is to enter a series of substitution values for the interest rate.

a. Make the **Home Loan** worksheet active, click **cell D4**, type **4** and then press **Ctrl+Enter**. Cell D4 is the first cell containing a substitution value. Make sure cell D4 is still the active cell.

b. Click **Fill** in the Editing group on the Home tab and select **Series**. The Series dialog box opens.

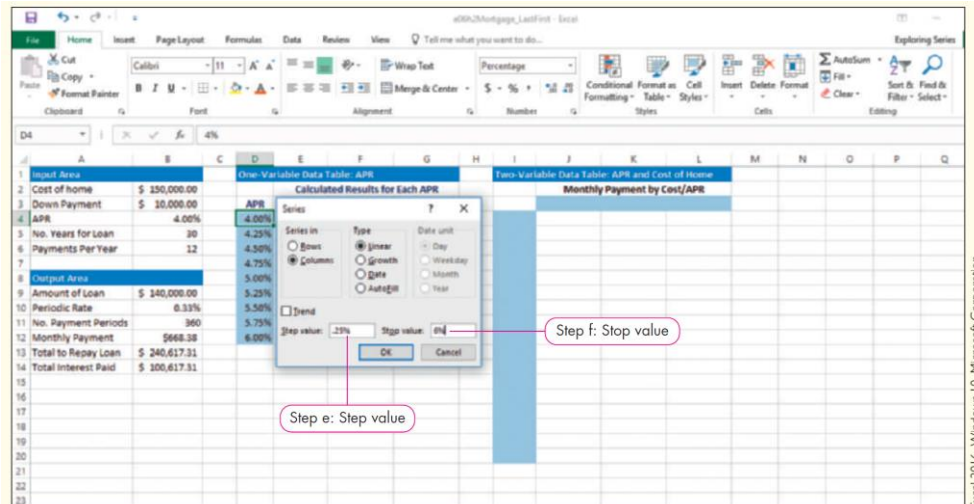
c. Click **Columns**.

You changed the *Series in* option to Columns because you want the series of substitution values listed vertically in column D.

d. Delete the existing value in the Step value box and type **0.25%**.

e. Type **6%** in the Stop value box and click **OK**. Excel fills in the series of values.

TROUBLESHOOTING: If you forget to type the decimal point in Step e and/or the percent sign in Steps e or f, the series will be incorrect. If this happens, click Undo and repeat Steps c through f.



Step 2: ADD FORMULAS TO A ONE-VARIABLE DATA TABLE AND CALCULATE RESULTS

In the next steps, you will enter references to the monthly payment, total amount to repay the loan, and total interest formulas.

Then, you will complete the table to compare the results for different interest rates ranging from 4% to 6%.

a. Click **cell E3**, type **=B12**, and then press **Tab**.

You entered a reference to the original monthly payment formula. When the results of cell B12 change, they are reflected in cell E3.

b. Type **=B13** in **cell F3** and press **Tab**.

You entered a reference to the original total amount to repay the loan.

c. Type **=B14** in **cell G3** and press **Enter**.

You entered a reference to the original total interest paid.

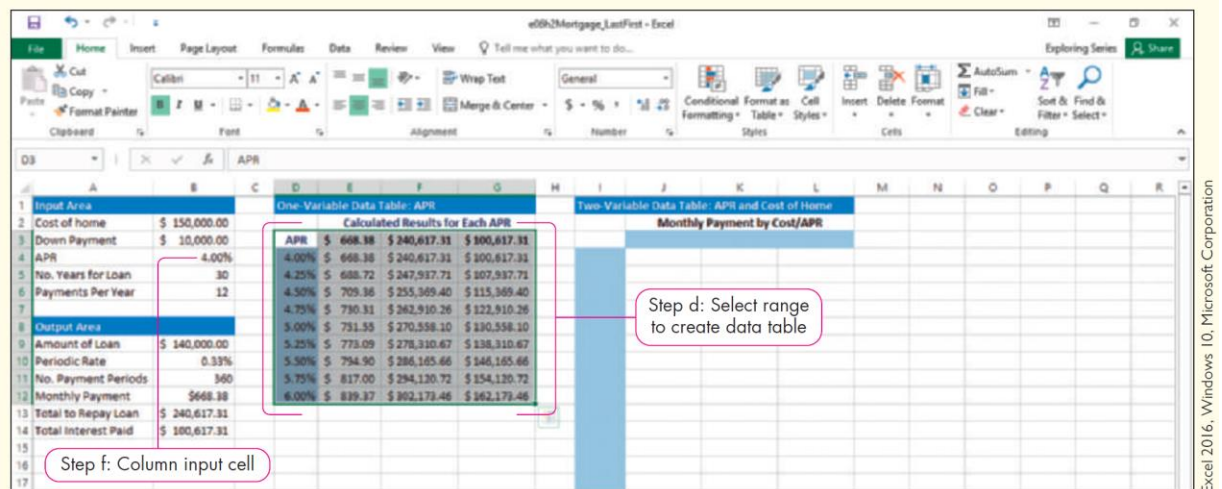
d. Select the **range D3:G12**.

You select the entire range of the data table, starting in the blank cell in the top-left corner. Note that you did not select the titles or headings in cells D1:G2.

e. Click the **Data** tab, click **What-If Analysis** in the Forecast group, and then select **Data Table**.

f. Click in the **Column input cell** box, click **cell B4**, and then click **OK**. Save the workbook. Because the substitution values are in a column, you reference cell B4 in the Column input

box. Excel inserts the TABLE array function in the empty result cells and substitutes the values in range D4:D12 individually for the original APR to calculate the respective monthly payments, total amounts, and total interest payments. The higher the APR, the higher the monthly payment, total amount to repay the loan, and total interest.



Step 3: FORMAT A ONE-VARIABLE DATA TABLE

You want to format the results to show dollar signs and to display rounded values to the nearest penny. In addition, you want to add column headings to provide more detail to the data table and add custom formats to the cells to appear as column headings.

a. Select the **range E4:G12**, click the **Home** tab, and then click **Accounting Number Format** in the Number group.

The values look more professional now that you have formatted them.

b. Click **cell D3**, type **APR**, and then press **Tab**.

Because cell D3 was empty, you can type the label directly in the cell without adding a custom format. Cell E3 should now be the active cell.

c. Click **Number Format** in the Number group on the Home tab.

d. Select **Custom** in the Category list, scroll up through the Type list, and select **General** in the list.

e. Select the existing text in the **Type box**, type **"Payment"** (make sure you include the quotation marks), and Click **OK**. The custom format is applied and Payment displays instead of =B12.

f. Repeat and adapt steps c through e to enter the following custom number formats:

"Total Repaid" for cell F3 and **"Total Interest"** for cell G3.

TROUBLESHOOTING: If you forget the quotation marks, the cell contents will contain a mix of numbers and characters. If this happens, open the Format Cells dialog box again and edit the contents of the Type box to display the text surrounded by quotation marks.

g. Center and bold the range E3:G3.

h. Save the workbook.

Steps d-f: Custom numbers formatted as headings

Input Area		One-Variable Data Table: APR				Two-Variable Data Table: APR and Cost of Home	
		Calculated Results for Each APR				Monthly Payment by Cost/APR	
		APR	Payment	Total Repaid	Total Interest		
Cost of home	\$ 150,000.00	4.00%	\$ 668.38	\$ 240,617.31	\$ 100,617.31		
Down Payment	\$ 10,000.00	4.25%	\$ 688.72	\$ 247,937.71	\$ 107,937.71		
APR	4.00%	4.50%	\$ 709.36	\$ 255,369.40	\$ 115,369.40		
No. Years for Loan	30	4.75%	\$ 730.31	\$ 262,910.26	\$ 122,910.26		
Payments Per Year	12	5.00%	\$ 751.55	\$ 270,558.10	\$ 130,558.10		
Output Area		5.25%	\$ 773.09	\$ 278,310.67	\$ 138,310.67		
Amount of Loan	\$ 140,000.00	5.50%	\$ 794.90	\$ 286,165.66	\$ 146,165.66		
Periodic Rate	0.33%	5.75%	\$ 817.00	\$ 294,120.72	\$ 154,120.72		
No. Payment Periods	360	6.00%	\$ 839.37	\$ 302,173.46	\$ 162,173.46		
Monthly Payment	\$668.38						
Total to Repay Loan	\$ 240,617.31						
Total Interest Paid	\$ 100,617.31						

Step d: Accounting Number Format applied

Step 4: SET UP TWO-VARIABLE SUBSTITUTION VALUES AND ADD A FORMULA TO THE DATA TABLE

Now you want to focus on how a combination of interest rates and different costs will affect just the monthly payment.

a. Enter **150000**, **175000**, and **200000** in the **range J3:L3**. Format these values with **Accounting Number Format**. Expand the column width if pound signs (#) appear.

b. Click **cell I4**, type **4%**, and then press **Ctrl+Enter**.

c. Click **Fill** in the Editing group on the Home tab, select **Series**, and then click **Columns**.

d. Replace the existing value in the Step value box with **0.25%**, type **8%** in the Stop value box, and then click **OK**.

e. Format the **range I4:I20** with two decimal places.

f. Click **cell I3**, type **=B12**, and then press **Ctrl+Enter**. You inserted the reference to the formula in the top-left cell of the two-variable data table. Save the workbook.

Step e: Percent Style Number Format applied

Step a: Values entered directly into cell

Step d: Enter substitution values

Input Area				One-Variable Data Table - APR				Two-Variable Data Table - APR and Cost of Home			
				Calculated Results for Each APR				Monthly Payment by Cost/APR			
				APR	Payment	Total Repaid	Total Interest	APR	\$150,000.00	\$175,000.00	\$200,000.00
Cost of home	\$	150,000.00		4.00%	\$ 668.38	\$240,617.31	\$100,617.31	4.00%			
Down Payment	\$	10,000.00		4.25%	\$ 688.72	\$247,937.71	\$107,937.71	4.25%			
APR		4.00%		4.50%	\$ 709.36	\$255,369.40	\$115,369.40	4.50%			
No. Years for Loan		30		4.75%	\$ 730.31	\$262,910.26	\$122,910.26	4.75%			
Payments Per Year		12		5.00%	\$ 751.55	\$270,558.10	\$130,558.10	5.00%			
Output Area				5.25%	\$ 773.09	\$278,310.67	\$138,310.67	5.25%			
Amount of Loan	\$	140,000.00		5.50%	\$ 794.90	\$286,165.66	\$146,165.66	5.50%			
Periodic Rate		0.33%		5.75%	\$ 817.00	\$294,120.72	\$154,120.72	5.75%			
No. Payment Periods		360		6.00%	\$ 839.37	\$302,173.46	\$162,173.46	6.00%			
Monthly Payment	\$	968.38		6.25%				6.25%			
Total to Repay Loan	\$	240,617.31		6.50%				6.50%			
Total Interest Paid	\$	100,617.31		6.75%				6.75%			
				7.00%				7.00%			
				7.25%				7.25%			
				7.50%				7.50%			
				7.75%				7.75%			
				8.00%				8.00%			

Step 5: CALCULATE RESULTS FOR A TWO-VARIABLE DATA TABLE

You complete the data table, format the monthly payment results, and apply a custom number format to the cell containing the formula reference so that it displays the text APR.

- Select the **range I3:L20**.
- Click the **Data** tab, click **What-If Analysis** in the Forecast group, and then select **Data Table**.
- Click **cell B2** to enter that cell reference in the Row input cell box. Because you entered the purchase price substitution values in the top row of the data table, you entered the reference to the cell containing the original cost in the Row input cell box.
- Click in the **Column input cell box**, click **cell B4**, and click **OK**. Because you entered the interest rate substitution values in the left column of the data table, you entered the reference to the cell containing the original APR variable in the Column input cell box.
- Click **cell I3** and apply a custom number format to display **APR**. Center and bold the contents in **cell I3**. Save the workbook.

The screenshot shows an Excel spreadsheet titled "e09h2Mortgage_LastFirst - Excel". The spreadsheet is divided into several sections:

- Input Area:** Contains input values for Cost of Home (\$150,000.00), Down Payment (\$10,000.00), APR (4.00%), No. Years for Loan (30), Payments Per Year (12), Amount of Loan (\$140,000.00), Periodic Rate (0.33%), No. Payment Periods (360), Monthly Payment (\$668.38), Total to Repay Loan (\$240,617.31), and Total Interest Paid (\$100,617.31).
- One-Variable Data Table: APR:** A table with columns for APR, Payment, Total Repaid, and Total Interest. It shows results for APR values from 4.00% to 6.00%.
- Two-Variable Data Table: APR and Cost of Home:** A table with columns for APR, Cost of Home, Payment, Total Repaid, and Total Interest. It shows results for APR values from 4.00% to 8.00% and Cost of Home values from \$150,000.00 to \$200,000.00.

Annotations on the spreadsheet indicate the following steps:

- Step c: Row input value:** Points to the input value for the Row input cell (B2) in the Data Table dialog box.
- Step d: Column input value:** Points to the input value for the Column input cell (B4) in the Data Table dialog box.
- Step e: Custom number format:** Points to the APR cell (I3) in the data table, where a custom number format is being applied.

Part (3)

Summary:

You want to use Goal Seek and Scenario Manager to perform additional what-if analyses with your mortgage data.

Step 1: DETERMINE OPTIMAL INPUT VALUES USING GOAL SEEK

Given the current interest rate with a 30-year mortgage and your planned down payment, you want to identify the most that you can afford and maintain a \$600.00 monthly payment. You will use Goal Seek to work backward from your goal to identify the ideal home purchase price.

- a. Click the **Data** tab.
- b. Click **What-If Analysis** in the Forecast group and select **Goal Seek**.
- c. Click **cell B12** to enter the cell reference for the monthly payment in the **Set cell box**. You indicated which cell contains the formula that produces the goal.
- d. Click in the **To value box** and type **600**. You want the monthly payment to be \$600.
- e. Click in the **By changing cell box** and click **cell B2**, the cell containing the cost of the home. Cell B2 is the cell whose value will be determined using the Goal Seek analysis tool.
- f. Click **OK**.
The Goal Seek Status dialog box opens, indicating that it reached the target monthly payment goal of \$600.

g. Click **OK** to accept the solution and to close the Goal Seek Status dialog box. Save the workbook.

To achieve a \$600 monthly mortgage payment, you need to purchase a home that costs up to \$135,676.74, instead of the original \$150,000, assuming the other variables (down payment, interest rate, and term of loan) stay the same.

Step h: Goal Seek results

Step g: Goal met

Step g: Goal Seek message

Goal Seek Status

Goal Seeking with Cell B12 found a solution.

Target value: 600

Current value: \$600.00

Excel 2016, Windows 10, Microsoft Corporation

Step 2: CREATE A SCENARIO

You want to use Scenario Manager to explore different scenarios. Your first scenario is a best-case scenario with these parameters: \$150,000 home, \$10,000 down payment, special reduced interest financing for 15 years.

a. Click **What-If Analysis** in the Forecast group on the Data tab, and then select **Scenario Manager**.

The Scenario Manager dialog box opens.

b. Click **Add**.

The Add Scenario dialog box opens so that you can assign a scenario name and select the changing cells.

c. Click in the **Scenario name box** and type **Best-Case Scenario**.

d. Delete existing contents in the *Changing cells* box and select the **range B2:B5**.

Excel enters this range in the Changing cells box. These are the variable cells cost of the home, down payment, and interest rate that will be changed in the various scenarios.

e. Ensure the Comment box displays your name and the date the scenario is created, such as “Created by Jason Davidson on 8/01/2018,” and click **OK**.
The Scenario Values dialog box opens so that you can enter the parameters for the scenario.

f. Type **150000** in the \$B\$2 box and press **Tab** twice to accept the current \$10,000 down payment.

g. Type **.0225** in the \$B\$4 box, press **Tab**, and then type **15** in the \$B\$5 box.

h. Click **OK** and click **Close**. Save the workbook.

While you could have kept the Scenario Values dialog box open to continue to the next step, you closed it so that you could save the workbook.

The screenshot shows an Excel spreadsheet with the following data:

Input Area		One-Variable Data Table: APR				Two-Variable Data Table: APR and Cost of Home			
		APR	Payment	Total Repaid	Total Interest	APR	\$150,000.00	\$175,000.00	\$200,000.00
2	Cost of home					4.00%	\$ 668.38	\$ 787.74	\$ 907.09
3	Down Payment					4.25%	\$ 688.72	\$ 811.70	\$ 934.69
4	APR	4.00%	\$ 600.00	\$ 216,000.00	\$ 90,323.26	4.50%	\$ 709.36	\$ 836.03	\$ 962.70
5	No. Years for loan	4.25%	\$ 618.25	\$ 222,571.45	\$ 96,894.71	4.75%	\$ 730.31	\$ 860.72	\$ 991.13
6	Payments Per Year	4.50%	\$ 636.79	\$ 229,242.82	\$ 103,566.07	5.00%	\$ 751.55	\$ 885.76	\$ 1,019.96
7		4.75%	\$ 655.59	\$ 236,012.18	\$ 110,335.44	5.25%	\$ 773.09	\$ 911.14	\$ 1,049.19
8		5.00%	\$ 674.66	\$ 242,877.58	\$ 117,200.83	5.50%	\$ 794.90	\$ 936.85	\$ 1,078.80
9	Amount of loan	5.25%	\$ 693.99	\$ 249,836.99	\$ 124,160.24	5.75%	\$ 817.00	\$ 962.90	\$ 1,108.79
10	Periodic Rate	5.50%	\$ 713.58	\$ 256,888.34	\$ 131,211.60	6.00%	\$ 839.37	\$ 989.26	\$ 1,139.15
11	No. Payment Periods	5.75%	\$ 733.42	\$ 264,029.53	\$ 138,352.79	6.25%	\$ 862.00	\$ 1,015.93	\$ 1,169.86
12	Monthly Payment	6.00%	\$ 753.50	\$ 271,258.41	\$ 145,581.66	6.50%	\$ 884.90	\$ 1,042.91	\$ 1,200.93
13	Total to Repay Loan					6.75%	\$ 908.04	\$ 1,070.19	\$ 1,232.34
14	Total Interest Paid					7.00%	\$ 931.42	\$ 1,097.75	\$ 1,264.07
15						7.25%	\$ 955.05	\$ 1,125.59	\$ 1,296.13
16						7.50%	\$ 978.90	\$ 1,153.70	\$ 1,328.51
17						7.75%	\$ 1,002.98	\$ 1,182.08	\$ 1,361.18
18						8.00%	\$ 1,027.27	\$ 1,210.71	\$ 1,394.15

The Scenario Values dialog box is open, showing the following values:

Cell Reference	Value
\$B\$2	150000
\$B\$4	.0225
\$B\$5	15

Callouts in the image indicate:

- Step f: Changing cells (points to the input area)
- Steps f-g: Best-Case Scenario Values for changing cells (points to the Scenario Values dialog box)
- Step b: Click to add another scenario (points to the Add button in the Scenario Values dialog box)

Step 3: ADD SCENARIOS

You will add two more scenarios: a worst-case scenario and a most likely scenario. In the worst-case scenario, you assume you will have to settle for a higher down payment, higher interest rate, and a longer loan period. In the most likely scenario, you will enter values that are between those in the other two scenarios.

a. Click **What-If Analysis** in the Forecast group and select **Scenario Manager**.

b. Click **Add**, type **Worst-Case Scenario**, and then click **OK**.

The *Changing cells* box displays B2:B5, the range you selected for the first scenario.

c. Type the following values in the respective changing cells boxes:

Changing Cell Box	Value
\$B\$2	150000
\$B\$3	45000
\$B\$	6%
\$B\$5	45

For the cell \$B\$4 box, you can enter the value as a percentage (6%) or as a decimal equivalent (0.06).

d. Click **Add**.

e. Type **Most Likely Scenario** and click **OK** in the Add Scenario dialog box.

f. Type the following values in the respective changing cells boxes:

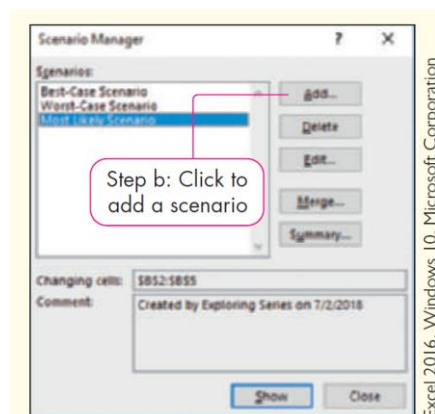
Changing Cell Box	Value
\$B\$2	150000
\$B\$3	30000
\$B\$4	4.25%
\$B\$5	30

g. Click **OK**.

The Scenario Manager dialog box lists the three scenarios you created.

TROUBLESHOOTING: If you believe you made any data entry errors, or if you want to double-check your values, select a scenario and click Edit. You can then change values in the Edit Scenario dialog box and click OK.

h. Click Close to close the Scenario Manager dialog box. Save the workbook.



Step 4: GENERATE A SCENARIO SUMMARY REPORT

You want to generate a scenario summary report to compare the three home loan scenarios you created.

- a. Click **What-If Analysis** in the Forecast group and select **Scenario Manager**.
- b. Click **Summary**. Excel may select a range within a data table.
- c. Select the **range B12:B14** to enter it in the Result cells box and click **OK**.

Excel generates the summary on a new worksheet named Scenario Summary. The report details the changes in payment amounts based on the variable inputs. You need to make a few deletions and add descriptive labels.

- d. Delete the following:

- Column A
- Row 1
- Current Values column
- Notes in the **range A13:A15**

- e. Enter descriptive labels in the following cells:

- **Cost of Home** in cell A5
- **Down Payment** in cell A6
- **APR** in cell A7
- **Years** in cell A8
- **Monthly Payment** in cell A10
- **Total to Repay Loan** in cell A11
- **Total Interest Paid** in cell A12

The labels describe data contained in each row. Now you can delete column B, which displays the cell references. Note, if range names are used with Scenario Manager, they will appear in the scenario summary report.

f. Delete column B and increase the width of column A.

Results based on the scenario summary report:

- ❖ The Worst-Case scenario provides the lowest payment but has the highest interest charges and longest repayment duration.
- ❖ The Best-Case scenario has the highest payment but the shortest repayment duration and lowest interest charges
- ❖ The Most-Likely scenario offers a monthly payment in-between Best and Worst-Case options while offering lower interest charges and a reduced repayment duration compared to the Worst-Case scenario

g. Save the workbook.

TIP: SCENARIO WORKSHEETS: Each time you generate a summary, Excel inserts another Scenario Summary worksheet. You can delete a summary worksheet if you no longer need the data.

Step e: Descriptive labels	<table border="1"> <thead> <tr> <th colspan="5" data-bbox="618 1092 821 1100">Scenario Summary</th></tr> <tr> <th data-bbox="618 1100 821 1104"></th><th colspan="2" data-bbox="821 1100 972 1104">Best-Case Scenario</th><th colspan="2" data-bbox="972 1100 1123 1104">Worst-Case Scenario</th></tr> <tr> <th data-bbox="618 1104 821 1110"></th><th colspan="2" data-bbox="821 1104 972 1110"></th><th colspan="2" data-bbox="972 1104 1123 1110"></th></tr> <tr> <th data-bbox="618 1110 821 1117">Changing Cells:</th><th colspan="2" data-bbox="821 1110 972 1117"></th><th colspan="2" data-bbox="972 1110 1123 1117">Most Likely Scenario</th></tr> </thead> <tbody> <tr> <td data-bbox="618 1117 821 1121">Cost of Home</td><td data-bbox="821 1117 844 1121">\$</td><td data-bbox="844 1117 972 1121">150,000.00</td><td data-bbox="972 1117 993 1121">\$</td><td data-bbox="993 1117 1123 1121">150,000.00</td></tr> <tr> <td data-bbox="618 1121 821 1127">Down Payment</td><td data-bbox="821 1121 844 1127">\$</td><td data-bbox="844 1121 972 1127">10,000.00</td><td data-bbox="972 1121 993 1127">\$</td><td data-bbox="993 1121 1123 1127">43,000.00</td></tr> <tr> <td data-bbox="618 1127 821 1134">APR</td><td data-bbox="821 1127 844 1134"></td><td data-bbox="844 1127 972 1134">2.25%</td><td data-bbox="972 1127 993 1134"></td><td data-bbox="993 1127 1123 1134">6.00%</td></tr> <tr> <td data-bbox="618 1134 821 1140">Years</td><td data-bbox="821 1134 844 1140"></td><td data-bbox="844 1134 972 1140">15</td><td data-bbox="972 1134 993 1140"></td><td data-bbox="993 1134 1123 1140">45</td></tr> <tr> <td data-bbox="618 1140 821 1144">Result Cells:</td><td data-bbox="821 1140 844 1144"></td><td data-bbox="844 1140 972 1144"></td><td data-bbox="972 1140 993 1144"></td><td data-bbox="993 1140 1123 1144"></td></tr> <tr> <td data-bbox="618 1144 821 1150">Monthly Payment</td><td data-bbox="821 1144 844 1150"></td><td data-bbox="844 1144 972 1150">\$917.12</td><td data-bbox="972 1144 993 1150"></td><td data-bbox="993 1144 1123 1150">\$963.10</td></tr> <tr> <td data-bbox="618 1150 821 1157">Total to Repay Loan</td><td data-bbox="821 1150 844 1157">\$</td><td data-bbox="844 1150 972 1157">165,081.36</td><td data-bbox="972 1150 993 1157">\$</td><td data-bbox="993 1150 1123 1157">304,073.35</td></tr> <tr> <td data-bbox="618 1157 821 1161">Total Interest Paid</td><td data-bbox="821 1157 844 1161">\$</td><td data-bbox="844 1157 972 1161">25,081.36</td><td data-bbox="972 1157 993 1161">\$</td><td data-bbox="993 1157 1123 1161">199,073.35</td></tr> </tbody> </table>					Scenario Summary						Best-Case Scenario		Worst-Case Scenario							Changing Cells:			Most Likely Scenario		Cost of Home	\$	150,000.00	\$	150,000.00	Down Payment	\$	10,000.00	\$	43,000.00	APR		2.25%		6.00%	Years		15		45	Result Cells:					Monthly Payment		\$917.12		\$963.10	Total to Repay Loan	\$	165,081.36	\$	304,073.35	Total Interest Paid	\$	25,081.36	\$	199,073.35
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Step c: Scenario results																																																																	

Excel 2016, Windows 10,
Microsoft Corporation

Save the workbook.

Keep a record of this file as **e06h1Mortgage_LastNme_FirstName**.

Please submit this file on the blackboard (Chapter 6, hands-on 1).